



Comprehensive Multi-Parametric Assessment of Mitochondrial Functional Changes in Chemo Resistant MDA-MB231

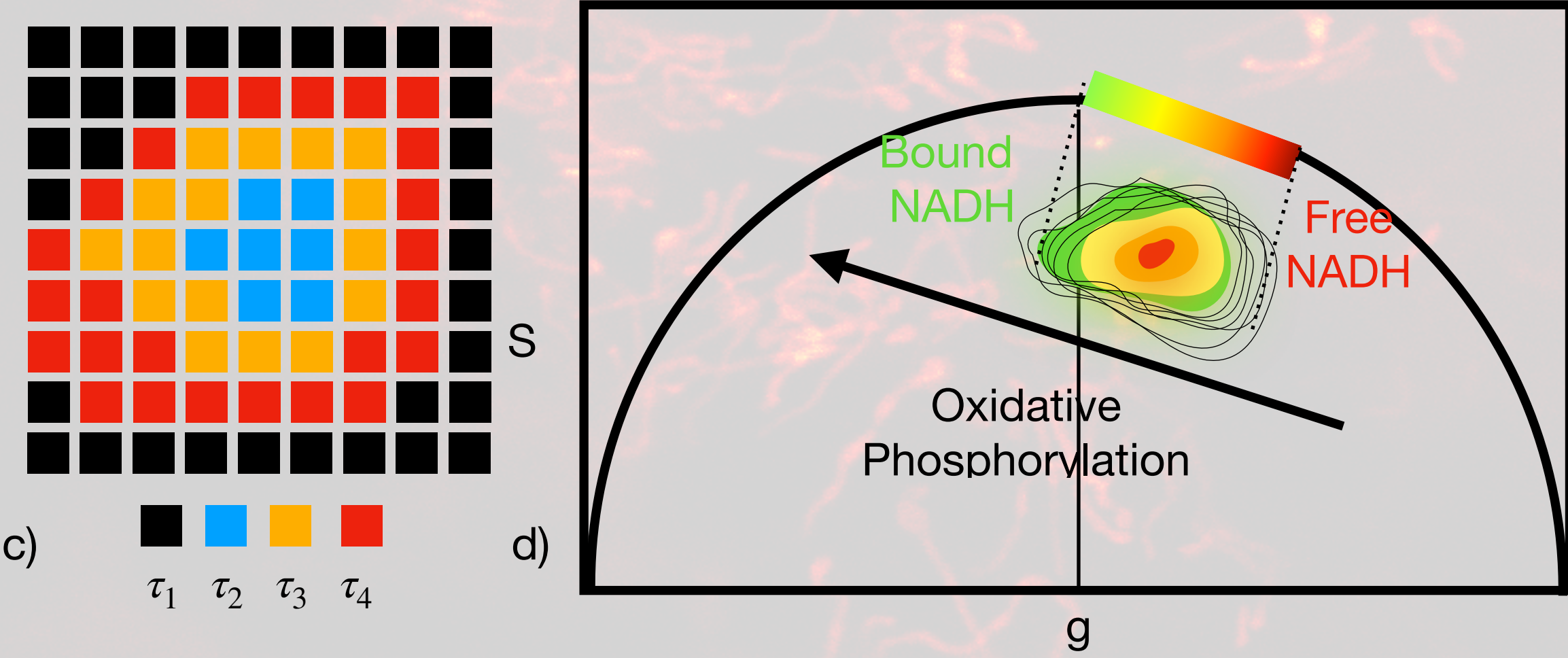
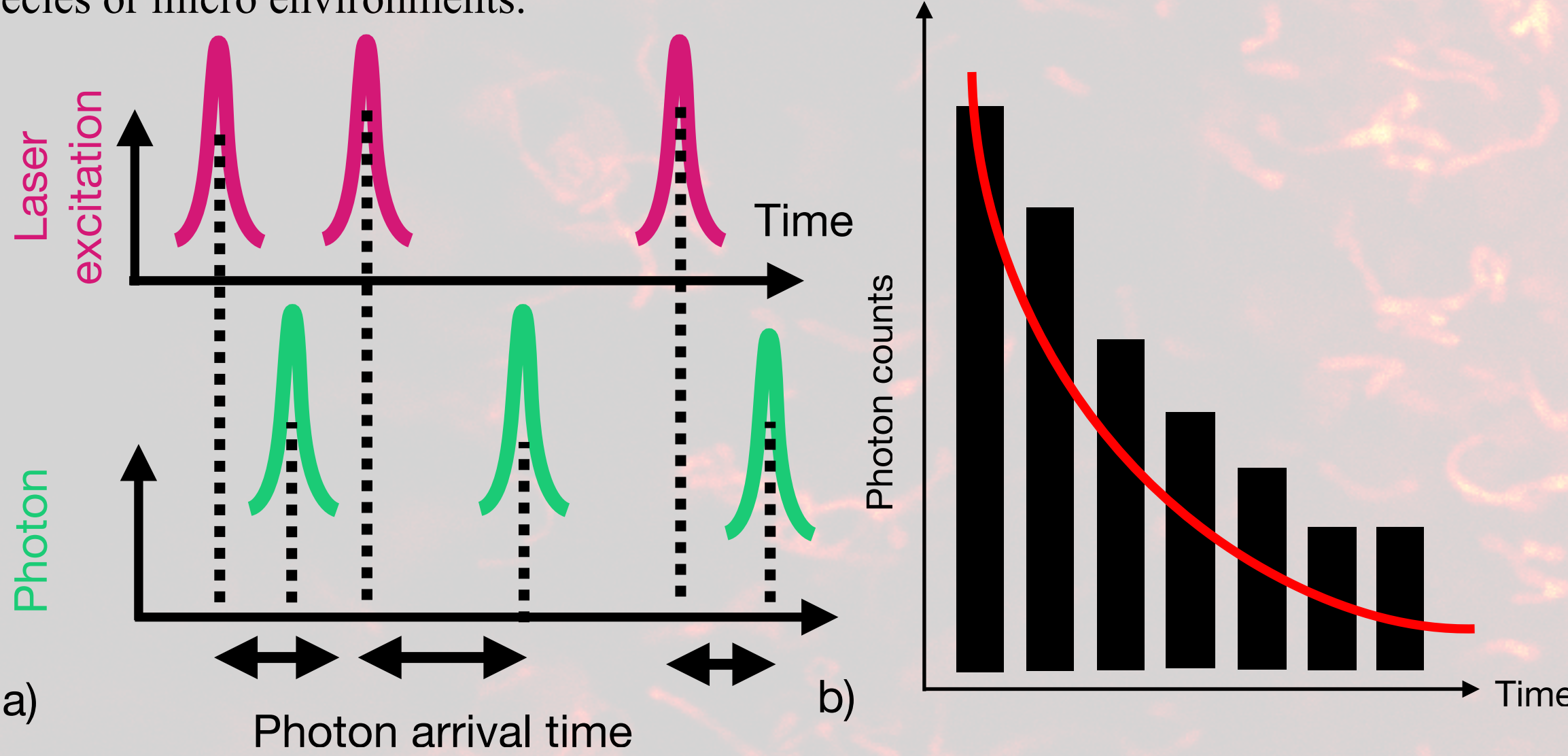
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Abstract

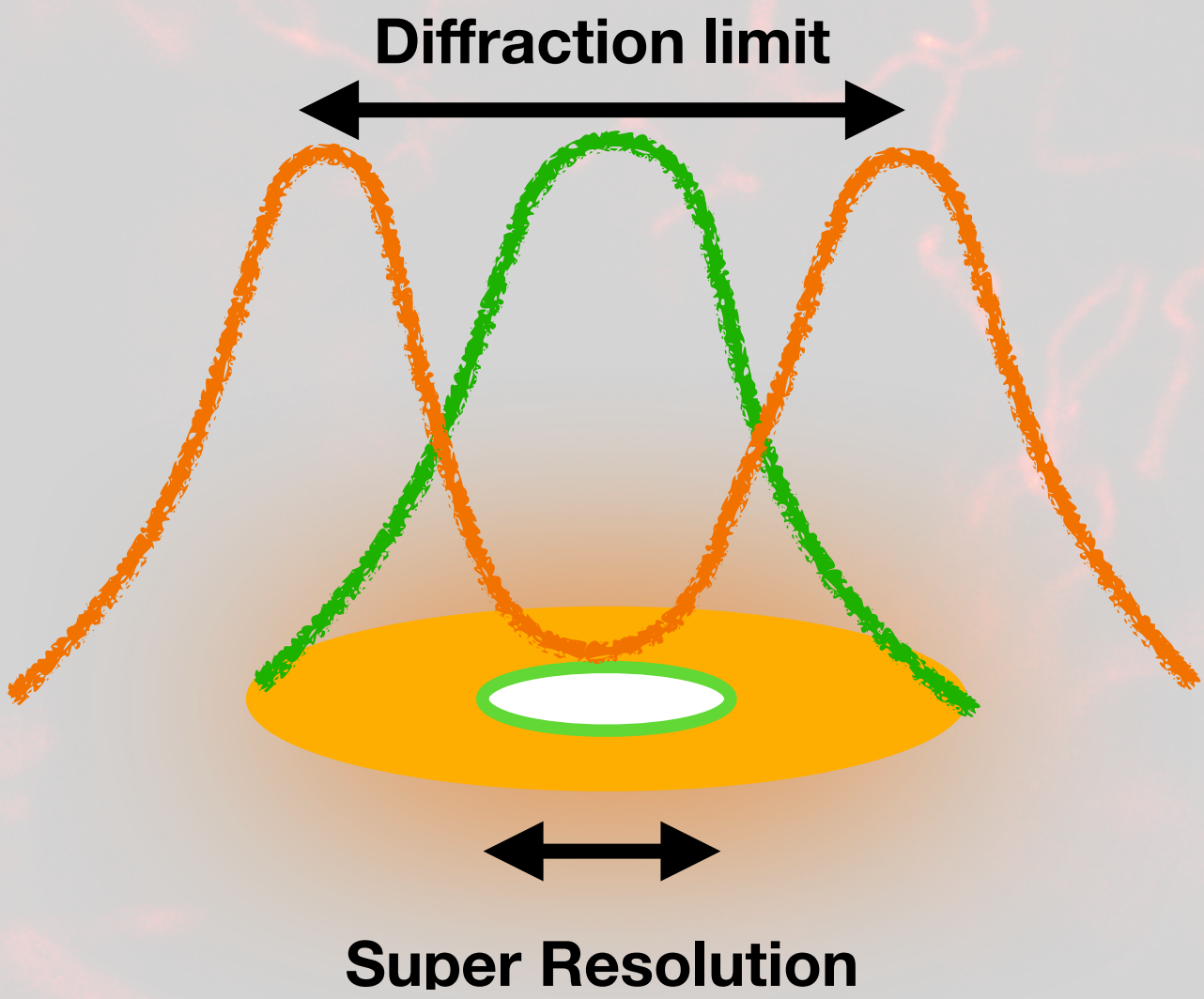
This study examines mitochondrial adaptations in chemoresistant breast cancer cells using multiphoton FLIM with the phasor approach and stimulated emission depletion microscopy. These techniques enable non-invasive, quantitative assessment of metabolic changes at the organelle level. MDA-MB-231 cells were analyzed before and after chemoresistance development, revealing distinct mitochondrial changes associated with drug adaptation. FLIM-based analysis demonstrated alterations in redox balance and biosynthetic activity, confirming mitochondria's role in sustaining chemoresistant phenotypes. By integrating FLIM and spectral imaging, this study provides a comprehensive multi-parametric assessment of mitochondrial metabolism, emphasizing organelle-specific metabolic profiling as a potential tool for identifying therapeutic targets in chemoresistant cancer.

Background & Introduction

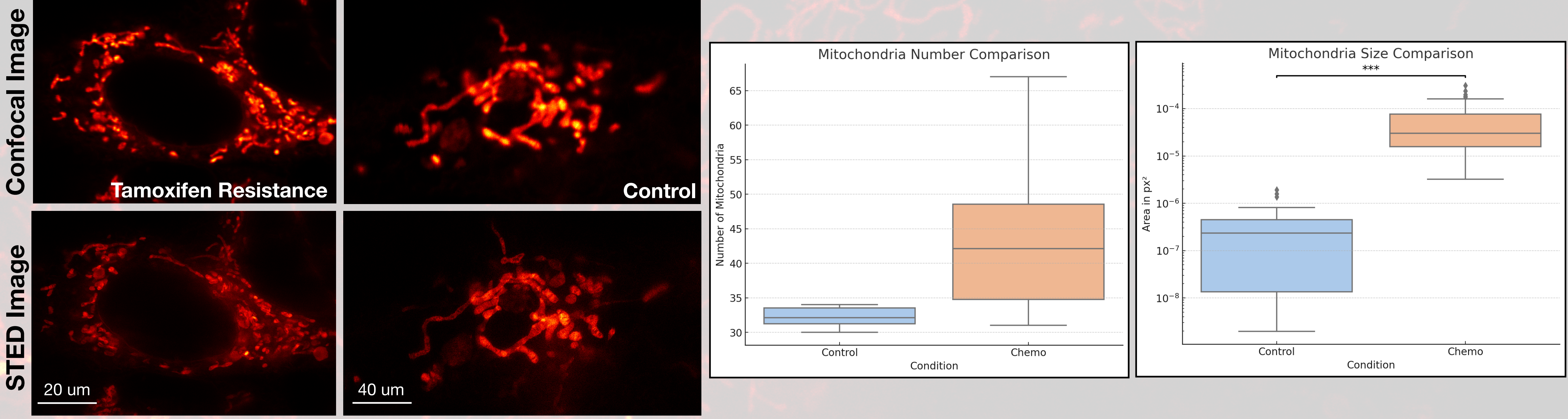
Phasor FLIM (Fluorescence Lifetime Imaging Microscopy) is an imaging technique that measures the lifetime of fluorophores, or the time they spend in the excited state before emitting light. Unlike traditional fitting methods, phasor analysis transforms the fluorescence decay at each pixel into a point on a 2D plot, providing a model-free, fast, and intuitive way to visualize lifetime variations. Each lifetime component maps to a unique location on the phasor plot, allowing for the separation of different molecular species or micro environments.



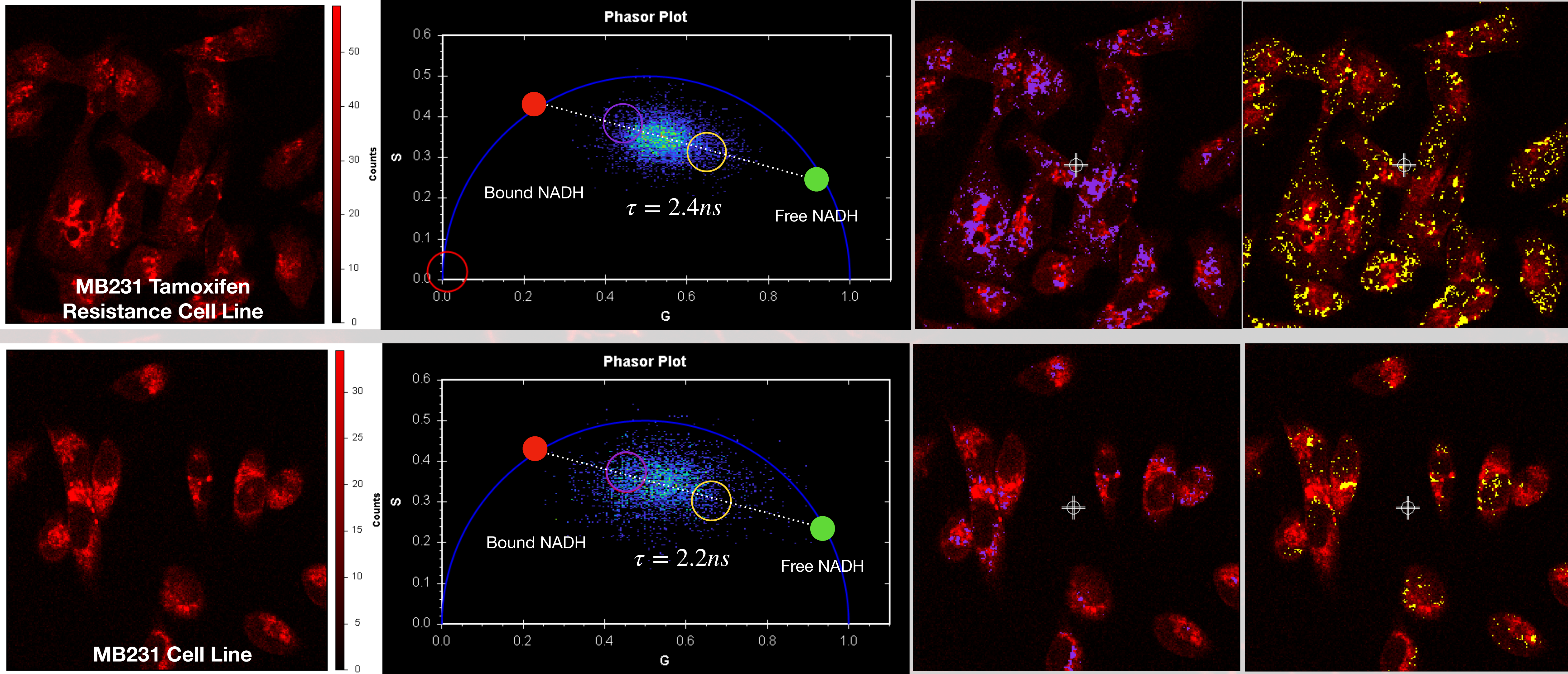
Stimulated Emission Depletion Microscopy



Fluorescence Life Time Imaging (FLIM)



Fluorescence Life Time Imaging (FLIM)



Conclusion & Acknowledgements

The significantly larger mitochondria in the chemo group could indicate that cells are shifting toward fusion and survival, rather than apoptosis. This morphological adaptation might be an early indicator of resistance development, allowing cells to withstand treatment and evade death pathways. Fused mitochondria have been shown to protect against apoptosis (programmed cell death), a major pathway by which chemotherapy aims to kill cancer cells. Enlarged mitochondria are often more resistant to mitochondrial outer membrane permeabilization (MOMP), which is a key step in apoptosis. Dysfunctional but persistent mitochondria can sustain low levels of ATP production, allowing cancer cells to survive under drug-induced stress. Overall, we conclude that tamoxifen-conditioned MDA-MB-231 cells exhibit enlarged mitochondria and a shifted glycolytic metabolic state compared to their control counterparts, suggesting tamoxifen-induced mitochondrial remodeling and metabolic reprogramming.

I thank Dr. Michelle Digman for her continuous support throughout this study. I also extend my gratitude to the entire LFD team, as well as Dr. Adeela Syed and Dr. Kristofer Fertig, for making the STED imaging possible.